

THE EFFECT OF SI CONTENT ON THE MICROSTRUCTURE AND MECHANICAL PROPERTIES OF Al-1.2Mg-xSi-1.2Cu-0.6Mn CAST ALLOY

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Abstract

In this paper, the effect of Si content on the microstructure and mechanical properties of the Al-1.2Mg-xSi-1.2Cu-0.6Mn cast alloy was investigated. The study aimed to explore the optimal addition of Si element to improve the comprehensive properties of alloys. The microstructure, phase composition and fracture morphology of the alloy were determined through OM, SEM, EBSD and XRD analysis. The Si content in the alloy ranged from 0.48 to 2.4 wt.%. With the increase of Si content, the number of strengthening phases increases, which improves the comprehensive properties of the alloy. When the Si content is 0.8 wt.%, the eutectic Mg₂Si transforms from rod-shaped to larger block shaped, and the formation of coarse Mg₂Si phase limits the elongation of the alloy. When the Si content is 2.4 wt.%, fine Al₂Cu phases are present in this alloy and coexist with Al(Fe,Mn)Si phases, while an increase in the Si content appears as partially accompanied by an

incipient crystalline Si phase around the Al(Fe,Mn)Si phase. The alloy exhibits a maximum tensile strength, yield strength and elongation of 198.2 MPa, 101.2 MPa and 4.96%, respectively, with a hardness of 80.94 Hv. It consists of five alloy phases, mainly α-Al, Mg₂Si, Al(Fe, Mn)Si eutectic phase, Q-AlCuMgSi eutectic phase and the beginning crystalline silicon phase formed due to the increased Si content. The Si content can improve alloy strength, but there is some damage to the toughness of the alloy. Higher silicon alloy content results in the formation of fine Al₂Cu eutectic. This result makes it possible to achieve a higher level of strength with a reduced loss of ductility in the alloy.

Keywords: cast Al-Mg-Si-Cu-Mn alloy, Si content, microstructure, mechanical properties

Introduction

Conventional iron crash barriers are expensive, easy to corrode, high maintenance cost and high recycling cost after a traffic accident. In order to respond to energy-saving and emission reduction, resource conservation as the global trend, so the development of new anti-collision crash barrier materials is very necessary.^{1,2} The addition of alloying elements improves the strength and toughness of mainstream cast aluminum alloys in Al-Mg-Si-Cu casting. The alloy's properties can be strengthened through subsequent treatments. Widely used in construction, mechanical parts, electronic communications, weapons, transportation and

aviation and other fields. A highly effective way to achieve the best overall alloy performance is to adjust the composition and post-treatment process.^{3,4}

The Mg and Si elements are the main strengthening elements of Al-Mg-Si alloys, Si elements in the alloy can increase the tensile strength and hardness of the alloy, but will reduce the toughness of the alloy.⁵ Zhu's⁶ results suggest that the incorporation of Si in die-cast Al-Mg-Si alloys resulted in the formation of a regular rod-like eutectic Mg₂Si phase, which led to reduced morphological irregularities and significant improvements in mechanical properties. The maximum tensile strength recorded was 350.12 MPa. It should be noted that there may be variations in the formation of different secondary phases depending on the Mg/Si ratios. If the Mg/Si ratio was 1.7, which corresponds to the mass ratio of Mg/Si in Mg₂Si, the added Mg and Si elements were consumed to form the Mg₂Si phase.⁷ Lu⁸

found that Mg_2Si possesses a high melting temperature of 1085°C , a low density of $1.99 \times 10^3 \text{ kg m}^{-3}$, a high hardness of $4.5 \times 10^9 \text{ Nm}^{-2}$, a low coefficient of thermal expansion of $7.5 \times 10^{-6} \text{ K}^{-1}$ and a high modulus of elasticity of 120 GPa . The addition of Si to the alloys provided grain refinement while increasing the flowability of the alloy melt and improving the mechanical properties of the alloys.⁹ Increasing the Si content in Al–Mg–Si alloys usually results in better formability and higher mechanical strength properties by increasing the density of finely dispersed Mg_2Si precipitates. Maximum tensile strength was up to 212.31 MPa .¹⁰ An excess of Si will produce hard Si particles which, under the influence of external forces, will cut the matrix and cause stress concentration in the alloy, thereby adversely affecting its properties.^{11,12} At the same time, Si inclusions in the primary phase and dispersion are inevitable, and many other elements exist more or less as unintentional impurities. For example, during the melting process of aluminum alloys, it is difficult to avoid the impurity Fe. Fe and Si elements in the alloys form needle-like $\beta\text{-AlFeSi}$ phases, which adversely affect the performance of the alloy.¹³ The addition of Mn to the alloy in a moderate amount can promote the passivation of the acicular $\beta\text{-AlFeSi}$ phase and the transformation of the $\alpha\text{-AlFeMnSi}$ phase into nearly circular, which is also conducive to the homogeneous distribution of the main strengthening phase, Mg_2Si and the improvement of the overall mechanical properties of the alloy.¹⁴ However, when the Mn content is too high, a brittle Al_6Mn phase is easily formed, which reduces the alloy's elongation.¹⁵ Copper also has a significant strengthening effect on Al–Mg–Si–Cu alloys. Shabestari et al.¹⁶ have found that increasing the Cu content from 0.2 to 1.5 wt.% can increase the tensile strength of such alloys by about 25%. Therefore, rational design of the alloy composition plays an important role in improving the properties of Al–Mg–Si alloys.

In order to investigate the effect of compositional changes on the microstructural evolution and mechanical properties of Al–Mg–Si alloys, the present work was carried out by further adjusting the composition of Si elemental content to modulate different Mg/Si ratios. The microstructures of the precipitated phases at different alloy compositions were studied, and the aluminum alloys with the desired ductility and strength were also obtained by performance testing. Therefore, in this paper, the effects of different Si element additions on the as-cast organization and properties of Al–1.2Mg–xSi–1.2Cu–0.6Mn alloys have been investigated in order to establish the relationships between Si element compositions, microstructures and mechanical properties.

Experimental Materials and Test Methods

Alloy Melting and Casting Process

The tests were carried out using industrial ingots of pure aluminum (> 99.8%), industrial ingots of pure magnesium (> 99.9%) and ingots of Al–20Si, Al–20Mn, Al–60Cu

intermediate alloys as raw materials. The mold was used for melting and the different raw materials were preheated separately; then, the raw materials were melted in a pit-type resistance furnace and the mold was coated with graphite before preheating. Pure aluminum was added, and the solution temperature was set at 700°C . When the pure aluminum began to melt, Al–20Si, Al–20Mn and Al–60Cu were added, and when the temperature was 670°C , pure Mg ingots were added, pressed into the melt and stirred slowly to minimize the loss of Mg by combustion and to ensure the accuracy of the composition. When the temperature was again raised to 700°C , weighed C_2Cl_6 was wrapped in aluminum foil and pressed into the melt using a hollow graphite rod, and the temperature was maintained at 700°C . The melt was allowed to stand for 10 min, after which the slag was removed. The molten metal was poured into a metal mold preheated to 150°C , and the pouring rate of the molten metal was controlled during the pouring process in order to fill the molten metal smoothly, prevent turbulence and reduce the occurrence of defects. Cooling was done by air cooling, and the ingot size of the casting using the steel mold was $200 \text{ mm} \times 40 \text{ mm} \times 120 \text{ mm}$. The alloy compositions have been determined by inductively coupled plasma spectroscopy, which is shown in Table 1. The simple casting flow chart is shown in Figure 1a. The melting and casting process is the same for all the samples.

Microstructure Characterization and Performance E Testing of Alloys

The metallographic specimens were cut into $10 \text{ mm} \times 5 \text{ mm} \times 10 \text{ mm}$ blocks from the same position of different alloy ingots. The surface for metallographic examination was the central portion 10 mm from the bottom of the casting. The specimens were mechanically ground and polished according to the standard specimen grinding procedure and etched using Keller mixed acid reagent for 15–25 s. Mechanical properties were tested at room temperature (25°C) using MTS-Exceed-E45, each tensile specimen was a sheet specimen machined from the bottom of the casting by wire EDM. Tensile tester was used to determine the ultimate tensile strength of the alloy, and yield strength and elongation were measured using an attached extensometer. To further improve the reliability, accuracy and repeatability of the experimental data, the tensile property results of each group of three parallel specimens were averaged, and the surfaces and sides of the tensile specimens were also polished with sandpaper prior to tensile testing to minimize experimental error. The microstructure and fracture morphology were observed using a Nikon model MA200 optical microscope (OM) and a ZEISS-EVO-18 scanning electron microscope (SEM); elemental distributions and semi-quantitative analyses of phase composition were analyzed using GENESIS, EDAX Company and America Energy Spectroscopy (EDS) and RIGAKUTTRIII-18 KW model X-ray diffractometer

Table 1. Al-1.2Mg-xSi-1.2Cu-0.6Mn Alloy Elemental Composition (wt.%)

Alloy	Mg	Si	Mn	Cu	Al	Fe	Ca	Mg/Si
S0.5	1.18	2.41	0.61	1.19	Bal.	0.261	0.019	0.489626556
S1.0	1.19	1.23	0.62	1.20	Bal.	0.211	0.003	0.967479675
S1.5	1.17	0.83	0.62	1.19	Bal.	0.222	0.05	1.409638554
S2.0	1.17	0.63	0.62	1.19	Bal.	0.251	0.004	1.857142857
S2.5	1.18	0.52	0.64	1.20	Bal.	0.279	0.002	2.269230769

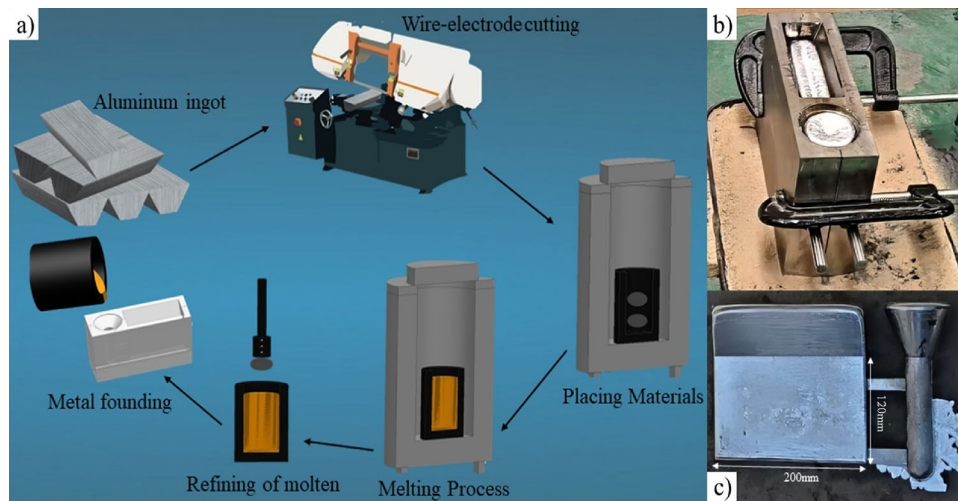


Figure 1. (a) Experimental casting process flow diagram; (b) ingot cooling; and (c) molding ingots.

(XRD) for phase analysis, and a Gemini 300 thermal field emission scanning electron microscope equipped with an electron backscatter diffraction (EBSD) probe (Oxford SYMMETRYS) was used to collect crystallographic orientation information of the samples. Hardness testing was carried out using a SHIMADZU HMV-G-FA micro Vickers hardness tester.

Results and Discussion

Microstructure

Figure 2a–e shows the microstructures of five alloys with different silicon contents at 50X. The gray area is the α -Al matrix, and the black area is the eutectic phase. The complete grain boundaries can be seen more clearly in the metallographies of the as-cast alloys after decay etching, and the black Mg_2Si is mainly mixed with eutectic phases such as $\text{Al}(\text{Fe},\text{Mn})\text{Si}$ and is alternately distributed in the intergranular space. These non-equilibrium phases roughly outline the grain boundaries, and black dot-like spots can also be seen within them. These spots are formed when the tetragonal eutectic phase Q-AlCuMgSi , i.e., the second phase $\theta\text{-Al}_2\text{Cu}$, corrodes in the cast alloy.¹⁷ Alloy in the as-cast state for the obvious dendritic organization, this is the alloy in the solidification process due to changes in

temperature field or composition ups and downs caused by the local alloy concentration is high, and caused by the growth of secondary, tertiary and other dendrite tendency to increase, the formation of dendritic spacing more and more fine dendritic crystals. The alloy exhibits significant compositional segregation, mainly by incipient dendritic α -Al and coarse intermetallic compound co-crystals between the dendrites. During solidification, the α -Al solid solution component of the alloy solidifies rapidly and cannot change according to the equilibrium solid phase line. At the eutectic temperature, a small amount of liquid remains and solidifies into the eutectic microstructure, resulting in non-equilibrium solidification. They are distributed on the dendrite and grain boundaries, and although they are sub-eutectic in organization, they appear as dissociated eutectics, i.e., the α -Al and α -Al substrates in the eutectic are connected, and the constituent eutectic compounds are presented individually as shown in Figure 2(f–j).^{18–20}

Figure 2f–j shows the grain boundary precipitation phases at 1000X for five alloys with different silicon contents. The intermetallic compounds that precipitated are labeled in Fig. 2f–j. The phases will be further determined using the EDS function of the scan. As is evident from Figure 2, the morphology of the non-equilibrium solidified phases between the α -Al grains of the alloys is also different. When the amount of Si is small, e.g., 0.48 wt.% for and 0.6

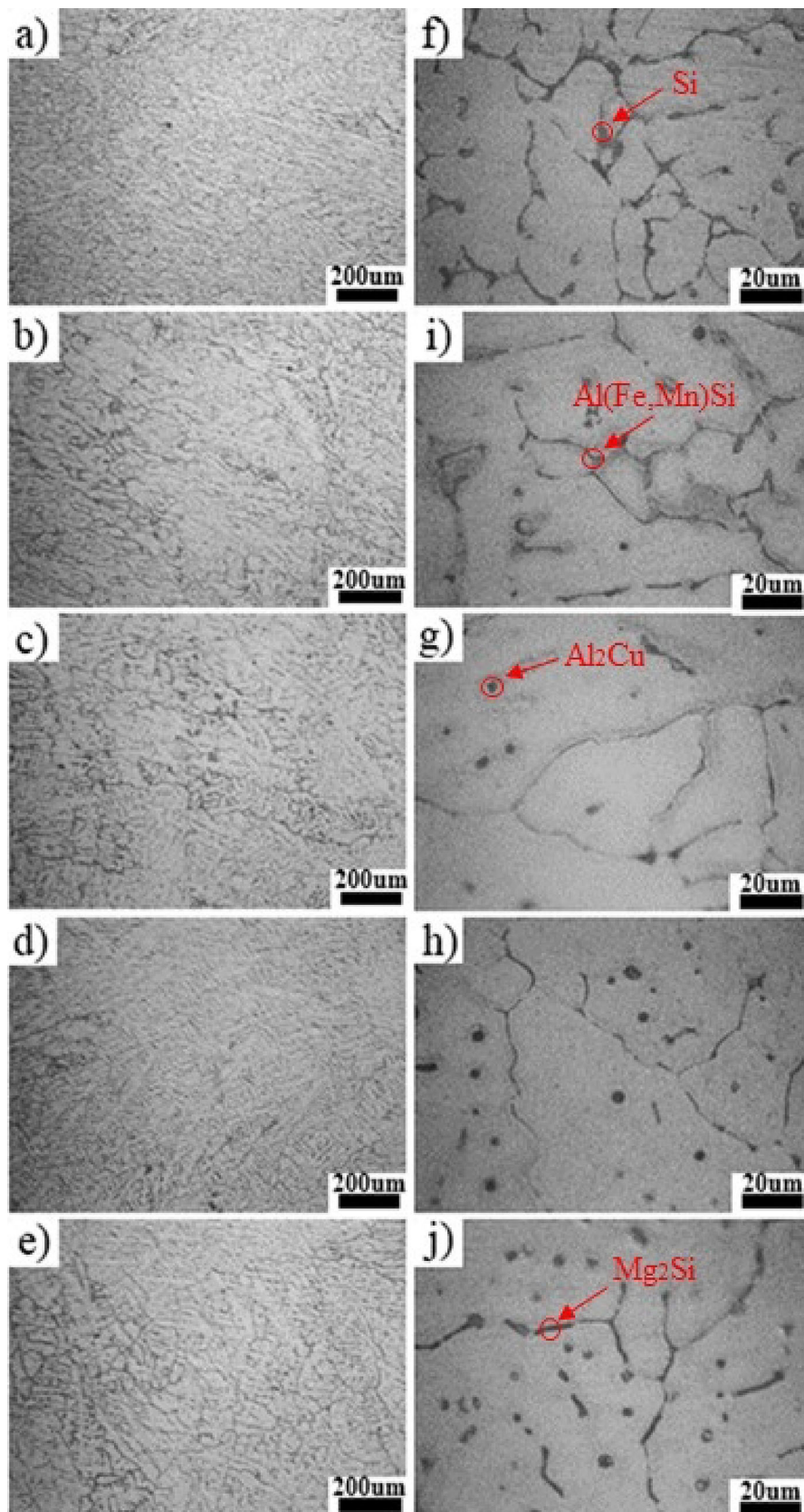


Figure 2. Metallographic micrographs of s0.5–s2.5 experimental alloys: (a, f) 2.4 Si; (b, g) 1.2 Si; (c, h) 0.8 Si; (d, h) 0.6 Si; (e, j) 0.48 Si.

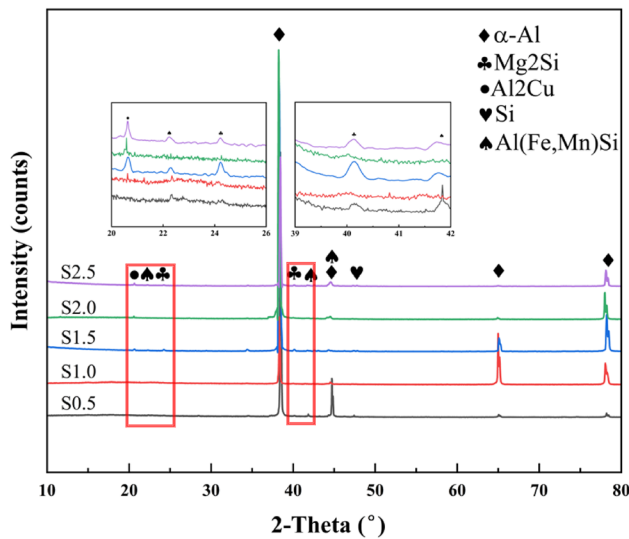


Figure 3. XRD patterns of Al-1.2Mg-xSi-1.2Cu-0.6Mn alloys.

wt.%, the phase structure at the grain boundaries is black elongated stripes. When the Si content is more than 1.2 wt.%, the morphology of the eutectic phase structure between the grains changes from elongated stripes to coarse granular eutectic and primordial crystalline Si phases.

The phase differences of five aluminum alloys with different compositions have been investigated by XRD, and the results are shown in Fig. 3. The pattern contains mainly diffraction peaks of α -Al matrix, Mg_2Si phase, $\text{Al}(\text{Fe,Mn})\text{Si}$ phase, incipient Si and Al_2Cu . When the amount of Si element is 2.4 wt.% and 1.2 wt.%, the Mg content is low and the Mg_2Si phase cannot be formed with all Si elements in the alloy, resulting in an excess of Si element in the alloy. Therefore, when the amount of Si is higher, diffraction peaks of eutectic Si phase are present in the XRD spectra, and the excess Si and Mn, Fe are more likely to form $\text{Al}(\text{Fe,Mn})\text{Si}$ eutectic phase as the amount of Si increases. The Al_2Cu phase was also detected in the XRD pattern, which is the second phase produced by the addition of Cu elements to the aluminum alloy, and no other Cu phases were produced in the alloy. The reason for the lower diffraction peaks of the Mg_2Si eutectic phase may be that the formation of the $\text{Al}(\text{Fe,Mn})\text{Si}$ eutectic phase and Q-AlCuMgSi phase in the alloy resulted in a lower content of the Si element, which allowed the generation of the Mg_2Si phase with a lower.

Figure 4 shows the as-cast SEM micrographs of the as-cast Al-1.2Mg-xSi-1.2Cu-0.6Mn alloys ($x = 2.4, 1.2, 0.8, 0.6, 0.48$) and the EDS analysis positions of the precipitated phases. Intermetallic compounds are the main strengthening phases in Al-Mg-Si as-cast alloys. Semi-quantitative analyses of the phases (labels in Figure 4) were carried out by EDS, and the results are shown in Table 2. Five types of phases were detected in the Al-1.2Mg-xSi-1.2Cu-0.6Mn

alloy, including incipient α -Al, incipient crystalline Si phase, $\text{Al}(\text{Fe,Mn})\text{Si}$ eutectic phase, Q-AlCuMgSi and Mg_2Si phase. The Mg_2Si phase, in which the elements have a low atomic number with respect to the Al substrate, is black on the gray Al substrate. The $\text{Al}(\text{Fe,Mn})\text{Si}$ phase, the primary Si phase and the Q phase have larger atomic numbers relative to the Al substrate and are white on the gray Al substrate. The Q phase is bright white, the $\text{Al}(\text{Fe,Mn})\text{Si}$ phase is off-white and the primary Si phase is lighter in color compared to the $\text{Al}(\text{Fe,Mn})\text{Si}$ phase.²¹

From Figure 5, it can be seen that there are obvious differences in the positions of the Mg, Mn, Si and Cu elements in the experimental alloys; Mg, Cu and Mn elements are enriched in part of the eutectic phase and are relatively uniformly distributed in the rest of the positions, while Si and Cu elements are enriched in the white phase with the surrounding region being the α -Al matrix. With an increase in Si content, the morphology of the precipitated intermetallic compounds changes, as does the distribution of each element's position and enrichment location to some extent. At low Si content, the Mg and Si elements become enriched with the precipitation of the Mg_2Si phase. The Cu elements are enriched in the precipitation position of the Al_2Cu phase, while the Mn elements are uniformly distributed in the α -Al matrix. This is due to the fact that the $\text{Al}(\text{Fe,Mn})\text{Si}$ quaternary eutectic phase does not precipitate. At high Si content, both Mg and Si elements are still enriched in the precipitation zone of the Mg_2Si phase. However, the enrichment of Mg elements is reduced due to the decrease in Mg_2Si phase precipitation at high Si content. The phase $\text{Al}(\text{Fe,Mn})\text{Si}$ is enriched with Si, Mn and a few impurity Fe elements, while Cu elements are distributed more uniformly throughout the alloy as the Al_2Cu phase precipitation changes.

The second phase boundary can be clearly identified from the IPF plot in Figure 6a. The Inverse pole-figure (IPF) plot mainly shows the random and preferential orientation of the grains, and it can be seen that the second phase of the S0.5 alloy boundary grows based on multiple heterogeneous nucleation sites. The mean kernel orientation deviation (KAM) is shown in Figure 6b, where the KAM values indicate the variation in local orientation errors due to geometrically necessary dislocations. The blue to red color bars in the plot indicate the KAM values from 0 to 4.45° . The higher the KAM value, the higher the dislocation density, which indicates the presence of a large amount of residual strain hindering grain development. It can be used to study the local lattice distortion, deformation and stored strain energy of the grains. The local orientation deviation obtained from the orientation image can be used to assess the amount of residual stress in the material. The figure shows the blue color covering the α -Al on the matrix and the green-yellow areas at the grain boundaries, indicating the growth of the second phase which leads to an increase in grain dislocation density and residual stresses.

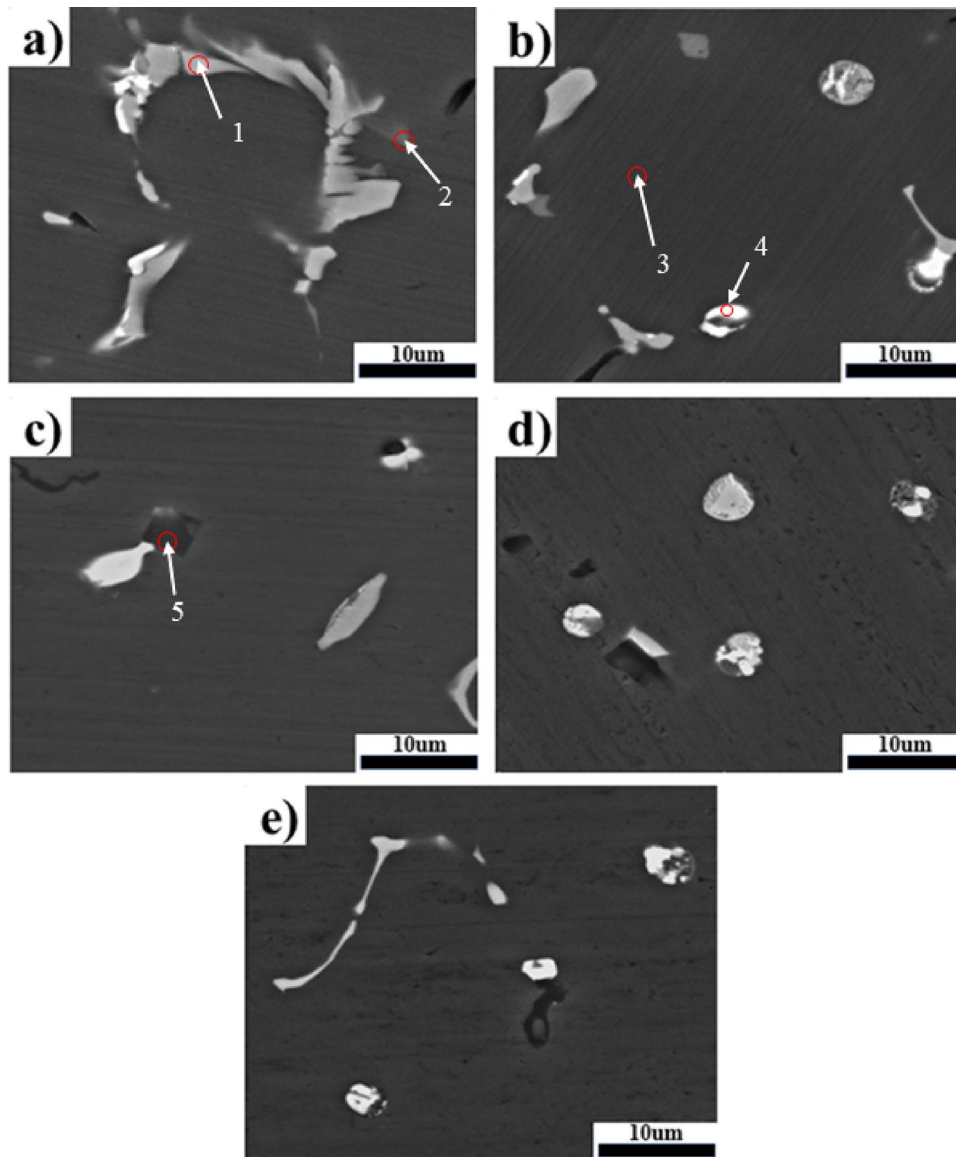


Figure 4. 5000x microstructure of an Al-1.2Mg-xSi-1.2Cu-0.6Mn alloy. (a) 2.4Si; (b) 1.2Si; (c) 0.8Si; (d) 0.6Si; and (e) 0.48Si.

Table 2. The EDS Composition of the Phase in Figure 4(At%)

Number	Mg	Si	Mn	Cu	Fe	Al	Phase
1	0.3	24.3	4.1	0.7	3.8	66.7	Al(Fe,Mn)Si
2	1.0	26.3	0	0.6	0	72.1	Si
3	0.5	0.9	0.2	0.2	1.5	98.2	α -Al
4	1.2	4.2	1.2	19.8	0.1	74.5	Al ₂ Cu
5	32.0	18.2	0	0.2	0	49.6	Mg ₂ Si

The EBSD phase diagram in Figure 6c shows that the Al(Fe,Mn)Si eutectic and Q phases are both mixed, but the difference can be clearly seen.

From the PF diagram in Figure 7, it can be found that the S0.5 alloy does not have more obvious grain orientation

changes in the as-cast state, and the grain orientation is usually closely related to the weave distribution, the polar plots of the above micro-regions were calculated as shown in Figure 7. The results show that the α -Al, Al(Fe,Mn)Si eutectic phase and Al₂Cu phase detected in the micro-region do not produce obvious weave changes, but it can be seen that the grain color changes in different regions are different, indicating that a certain concentration of crystalline orientation has been produced. The maximum orientation density of the Al matrix at the interface of the (100) grain surface has the extreme value of 59.71, which is larger in the microwave; and the grain orientation of the Al(Fe,Mn)Si and Al₂Cu phases is different from those in the microzone. Mn)Si and Al₂Cu phases have smaller microstructures with maximum orientation density extremes of 33.79 and 18.79 at the grain boundary, respectively.

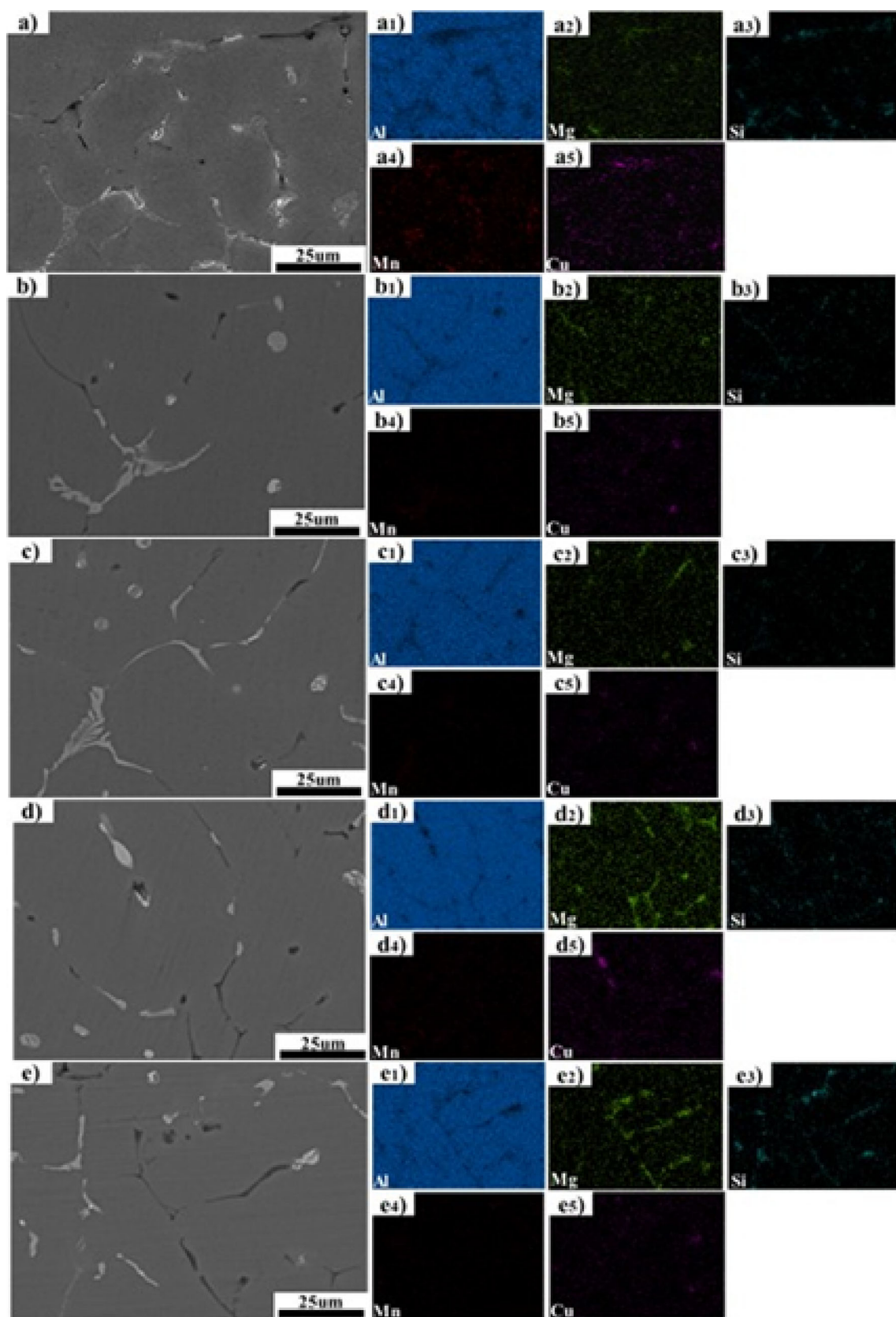


Figure 5. EDS surface scan of (a) s0.5, (b) s1.0, (c) s1.5, (d) s2.0, and (e) s2.5 alloys.

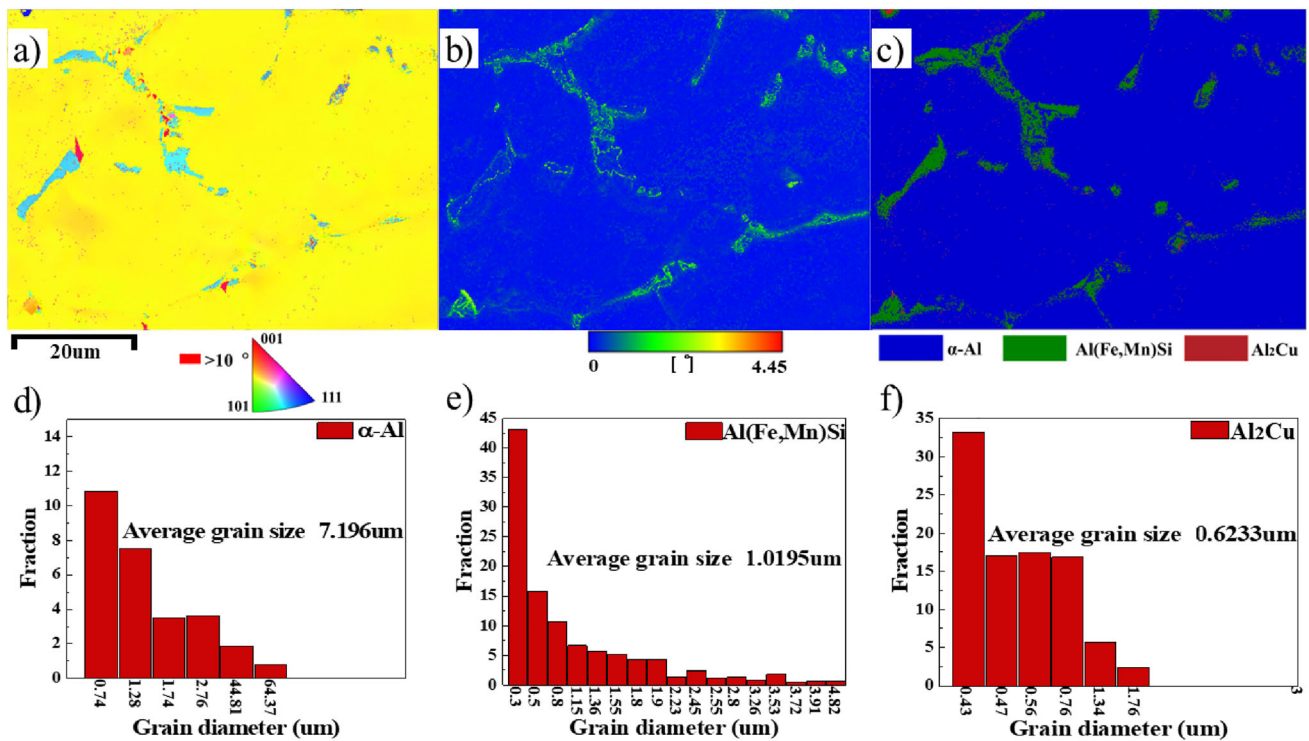


Figure 6. EBSD figures for S0.5 alloy: (a) reciprocal polar figure (IPF); (b) surface distribution diagram (KAM); (c) EBSD phase diagrams; and (d-f) grain size diagrams.

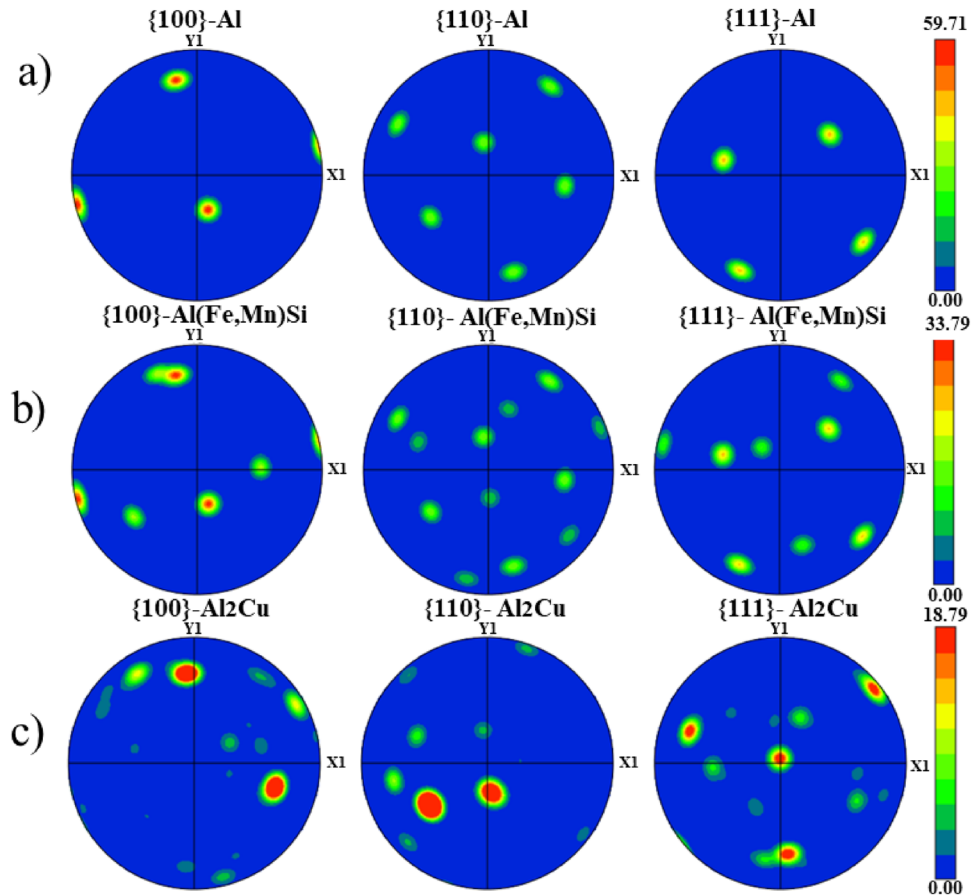


Figure 7. Pole figure of S0.5 alloys (PF): (a) α -Al; (b) Al(Fe,Mn)Si; and (c) Al₂Cu.

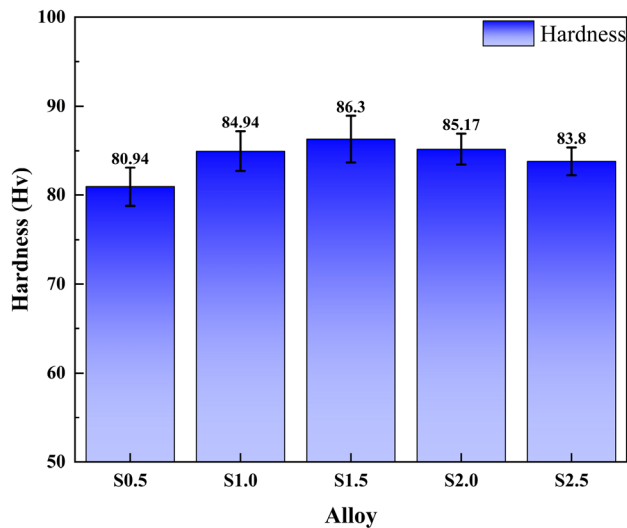


Figure 8. Hardness test plot of Al-1.2Mg-xSi-1.2Cu-0.6Mn alloy.

Mechanical Properties and Fracture Analysis

To investigate the effect on alloy properties, the experimental alloys with different Si contents were tested using Vickers hardness tester. The hardness results are shown in Figure 8. It is observed that the trend of hardness of the alloys shows an increasing and then decreasing trend with increasing Si content. This may be due to the fact that the increase in Si content from 0.5 to 1.5 wt.% alloys is due to the increase in Mg_2Si phase in the alloys, which leads to the increase in hardness of the alloys, whereas when the amount of Si is 2.4 wt.% and 1.2 wt.%, the alloys produce more $Al(Fe,Mn)Si$ phases at the grain boundaries and the larger size of the primary Si inclusions of the Al_2Cu phases. At the same time, the number of Mg_2Si phases decreases, accelerating the decreasing trend of alloy hardness.

Figure 9 shows the variation of tensile strength, yield strength and elongation of cast alloys with different amounts of Si; Figure 10 shows the variation of engineering stress-strain curves of cast alloys with different amounts of Si. The maximum yield strength and tensile strength of the alloys were 122.2 MPa and 209.7 MPa, respectively, when the amount of Si was 0.8 wt.%. When the amount of Si was increased from 0.48 to 0.8 wt.%, the yield strength and tensile strength of the alloys were increased, but the elongation decreased significantly. When the amount of Si was increased from 0.8 to 2.4 wt.%, the law is just the opposite, but the trend of tensile strength decrease of S0.5 alloy is not very obvious, and the addition of appropriate amount of Si can improve the strength of the alloy,²² and when the amount of Si is 0.48 wt.%. When the amount of Si is 0.48 wt.%, the amount of Mg_2Si phases formed in the alloy is less and the remaining amount of Si element is less, so that the Al_2Cu of this hard and brittle phase is formed more and coarser on the aluminum matrix

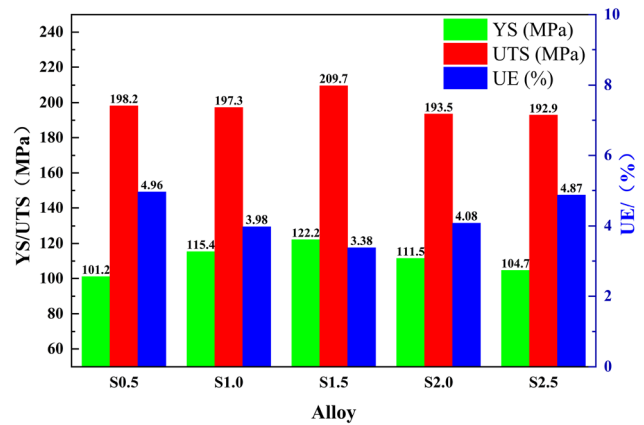


Figure 9. Tensile properties of Al-1.2Mg-xSi-1.2Cu-0.6Mn alloys at room temperature.

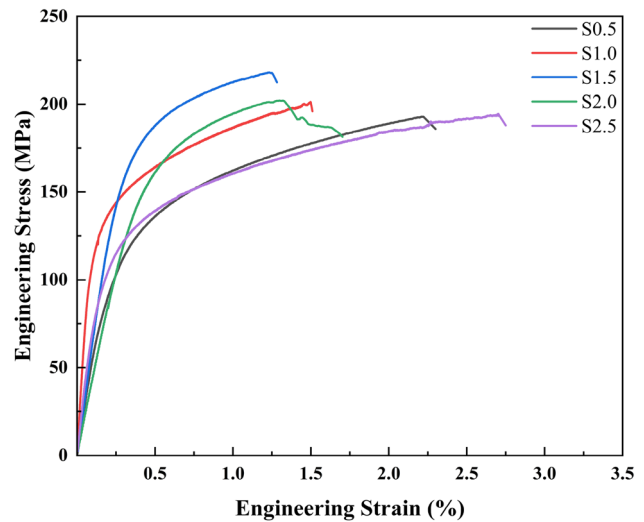


Figure 10. Engineering stress-strain diagrams for Al-1.2Mg-xSi-1.2Cu-0.6Mn alloys.

to produce the effect of cracking, so when the content of Si is 0.48 wt.%, the strength and toughness of the alloy are lower.^{23,24} From a microscopic point of view, the increase in strength of the alloy is due to the increase in the number of Mg_2Si phases in the alloy with increasing Si content, which improves the strength of the alloy, but the aggregation of coarse massive Mg_2Si phases tends to split the matrix and make the alloy less ductile. However, if the Si content is too high, the Mg_2Si phase in the matrix decreases and becomes finer, while more $Al(Fe,Mn)Si$ eutectic phase is produced, mixed with Al_2Cu phase and accompanied by part of the primary crystalline Si phase. The finer Al_2Cu phase with the Mg_2Si phases has a strengthening effect on the matrix, resulting in less toughness damage to the alloy, but more $Al(Fe,Mn)Si$ eutectic phase is prone to matrix cleavage^{25,26}.

In order to further investigate the effect of the Si content on the tensile strength of the Al-xMg-2.0Si-0.6Mn alloys, the fracture morphology of the fracture surfaces of the alloys

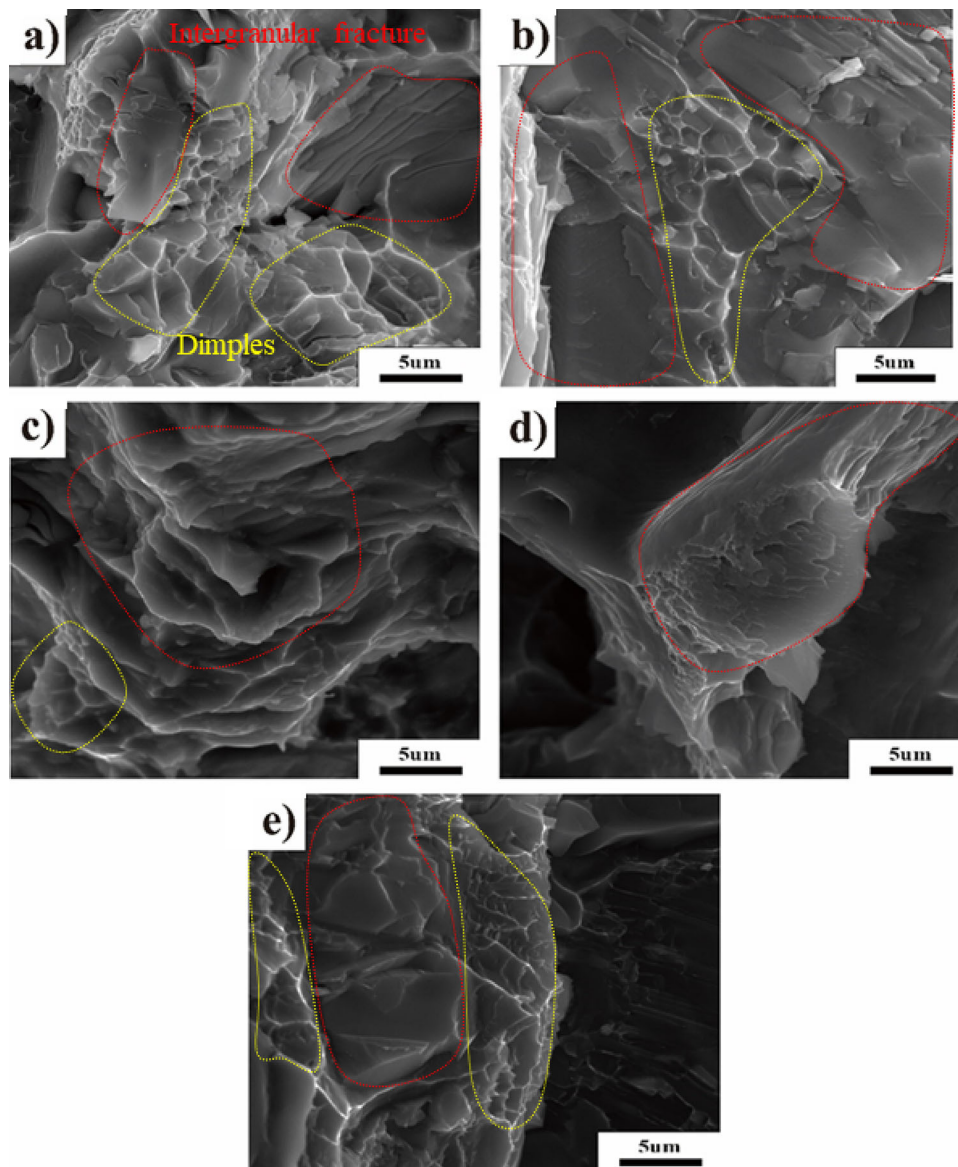


Figure 11. Fracture morphologies of Al-1.2Mg-xSi-1.2Cu-0.6Mn alloys at 6000x.

with different Si additions was analyzed by SEM, as shown in Figure 11. When the amount of Si was 0.5 wt.%, tearing edges and pits appeared in the alloy, indicating that the toughness of the alloy was better and the fracture mode was ductile fracture, and as the amount of Si increased, the S1.5 alloy appeared to have segregated fracture along the crystal in the surface fracture, with a large fracture surface, and the fracture mode was brittle fracture.²⁷ The number of fracture edges in the alloy increases when the amount of Si is 2.4 wt.%, indicating an increase in the toughness of the alloy.

As mentioned above, the precipitation phases of Al-Mg-Si alloys with the addition of Si are as follows in Figure 12. The results indicate that the grain boundary precipitation phases of Al-Mg-Si alloy with Si addition remain unchanged when the Si addition does not exceed 1.2%.

However, with an increase in Si content, the Mg/Si ratio becomes unfavorable for the formation of Mg₂Si phase, leading to a gradual decrease in the precipitation of Mg₂Si phase between grain boundaries. When the Si content was increased to 2.4 wt.%, the Si granular phase accompanied by Al(Fe,Mn)Si phase around the grain boundary gradually appeared with the increase of Si content, and at the same time, the Q phase gradually appeared with the trend of decreasing precipitation amount and the phase size became smaller, and at the same time, the Al(Fe,Mn)Si phase also gradually grew on the Q phase.

Conclusions

- (1) The types of alloy phases with different Si contents are basically unchanged, mainly α -Al, Mg₂Si, Al(Fe,Mn)Si eutectic phase,

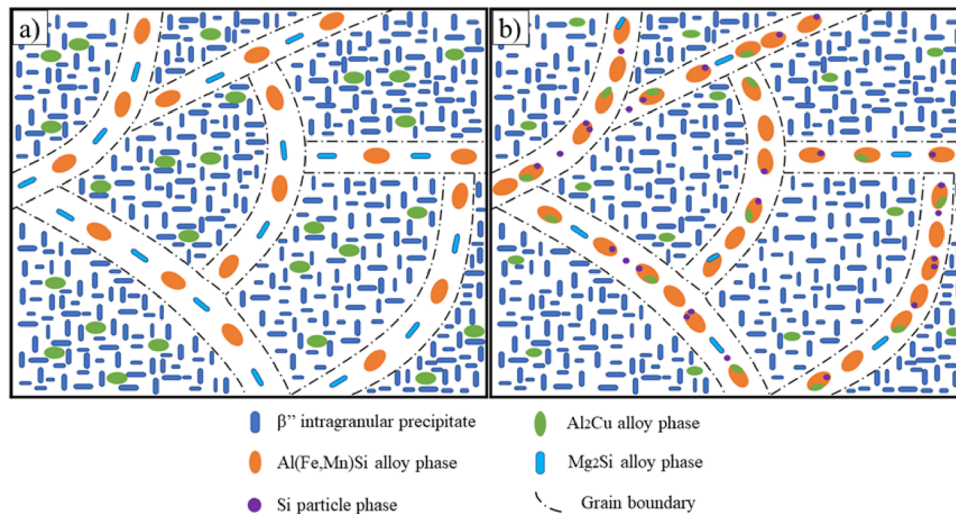


Figure 12. Schematic diagram of the precipitation phases of Al-1.2Mg-xSi-1.2Cu-0.6Mn alloy: (a) low Si and (b) high Si.

Q-AlCuMgSi eutectic phase, and some primary crystalline Si phase are formed when Si element is added in excess. As the Si content increases, the Al(Fe,Mn)Si eutectic phase precipitates more at the grain boundary and envelops the Q phase to some extent, and the Q phase has a finer grain size..

- (2) The mechanical properties of the alloys initially increased and then decreased with increasing Si content. At 0.8 wt.% Si, the maximum tensile strength, yield strength, elongation and hardness of the alloy were 209.7 MPa, 122.2 MPa, 3.38% and 86.3 Hv, respectively. The increase in the number of Mg₂Si phase and Q phase of the alloy improved the strength and hardness of the alloy, and the roughened massive Mg₂Si phase and the coarser Q phase, although they increased the hardness of the alloy, tended to cause matrix cleavage and reduced the elongation of the alloy
- (3) The AlCuMgSi eutectic phase can be used as a potential substrate for precipitating the Al(Fe,Mn)Si eutectic phase, which is easily mixed with the AlCuMgSi eutectic phase to be precipitated when the Si content is too much; the morphology and size of the Mg₂Si and AlCuMgSi eutectic phases have a greater effect on the mechanical properties of the alloy.

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